Visual Asset & Image Attachment Specification

Point-by-Point Blueprint: UI Screenshots, Photorealistic Prompts & System Diagrams

Executive Purpose: This document maps out all visual assets required for the 16 sections of the **AI Teacher Technical Documentation**. It categorizes every item as either **[REQUIRED]** (essential for evaluation rubrics) or **[OPTIONAL]** (aesthetic enhancement). For non-screenshot diagrams and educator portraits, it provides **photorealistic, non-AI-looking prompt specifications** with strict negative prompt guidance to ensure enterprise-grade quality suitable for hackathon judges and IEEE-style technical reports.

Dimension	Standards to Avoid the 'Generic AI Look'
UI Screenshots	Capture live from http://localhost:8000 using **Win + Shift + S** at 1920x1080 resolution. Keep Dark Mode consistent across all screens. Use subtle colored highlight boxes to draw attention to interactive elements.
Diagrams & Schemas	Export directly from Mermaid.js, Draw.io, or Graphviz as vector PNG/SVG with dark slate backgrounds (#0A0E1A). Avoid generic AI art generators for diagrams to eliminate misspelled labels or distorted boxes.
Educator Portraits	Prompts must mandate: *'DSLR 85mm portrait, f/1.8 lens, natural daylight, authentic skin pores/texture, realistic eyes with reflections, zero airbrushing/waxy skin, candid classroom background'*. Negative prompt: *'3D render, cartoon, plastic skin, CGI, distorted eyes, anime'*.
Priority Legend	**[REQUIRED - CRITICAL]**: Must be attached to pass the hackathon evaluation criteria. **[OPTIONAL - ENHANCEMENT]**: Supplementary illustration to achieve a flawless 100/100 visual presentation.

Master Visual Blueprint: Section-by-Section Mapping & Prompts

Sec & Status	Figure & Type	Visual Generation Prompt / UI Capture Target	Placement & Ready Caption
1. Problem **[REQUIRED]**	**Figure 1.1** **Infographic Diagram**	**Prompt (Midjourney / FLUX):** *'Flat vector minimalist corporate infographic comparing three education modes on dark slate background (#0A0E1A). Column 1: Static Video Lecture with red warning icon for passive learning. Column 2: Text Chatbot with yellow warning for wall-of-text fatigue. Column 3: AI Virtual Teacher with glowing emerald badge showing video + voice + adaptive whiteboard. Clean lines, McKinsey tech report style. --no 3D, cartoon, chaotic, blur'*	**Placement:** Below para 2. **Caption:** *Figure 1.1: Cognitive limitations of static video & text AI compared to interactive video tutoring.*
2. Solution **[REQUIRED]**	**Figure 2.1** **System Pipeline Flowchart**	**Prompt / Mermaid Code:** *'Professional enterprise software architecture flowchart on dark theme. Left-to-right sequential nodes: Ingestion (Textbooks/Topic) ➔ RAG Vector Store (ChromaDB) ➔ Pedagogy State Machine ➔ Multimodal Synthesis (Edge-TTS, Matplotlib, OpenCV) ➔ Reactive Video Classroom. Crisp geometric boxes, glowing gradient connecting arrows, clean sans-serif typography.'*	**Placement:** Below Section 2. **Caption:** *Figure 2.1: End-to-end architectural pipeline from course documents to interactive video delivery.*
3. Features (A) **[REQUIRED]**	**Figure 3.1** **UI Screenshot**	**Action:** Capture the **Source & Topic** tab (http://localhost:8000). Show: 1) Direct Topic input box, 2) Document Dropzone area, 3) Adaptive Time Selector (■ 5m, ■■ 20m, ■ 60m, ■ 7d), and 4) Teacher Persona selector cards. Add subtle red callout boxes around time and persona selectors.	**Placement:** Under Ingestion row. **Caption:** *Figure 3.1: Source ingestion interface with adaptive time budgeting and educator persona selectors.*
3. Features (B) **[REQUIRED]**	**Figure 3.2** **UI Screenshot**	**Action:** Capture the **Interactive Lesson** tab while running Ohm's Law or Chapter 2. Show the 2-column layout: Left 60% with Whiteboard circuit visual; Right 40% with the talking AI Teacher video player, multilingual language pills, and educator spoken script captions below.	**Placement:** Under Classroom row. **Caption:** *Figure 3.2: 2-column multimodal classroom featuring synchronized whiteboard diagrams and teacher video.*
3. Features (C) **[OPTIONAL]**	**Figure 3.3** **UI Screenshot**	**Action:** Click '■■ Flashcards' button in the lesson toolbar. Capture the active recall 3D flip card modal showing front question, flip button, and mastery counter. Inset a small preview of the downloaded StudyNotes_Ohm's_Law.md file.	**Placement:** Beside Retention row. **Caption:** *Figure 3.3: Interactive active recall flashcards modal and exported downloadable study notes guide.*
4. Architecture **[REQUIRED]**	**Figure 4.1** **Architectural Blueprint**	**Prompt / Mermaid / Draw.io:** *'Multi-tier enterprise cloud architecture diagram. Layer 1: Client React 18 SPA (HTML5 Range Streaming, Web Speech). Layer 2: FastAPI ASGI Server (REST routes, auth guard). Layer 3: Core AI Engines (Pedagogy State Graph, RAG, OpenCV Lip-Sync, Edge-TTS). Layer 4: Persistence (SQLite, ChromaDB) & Groq Cloud API. Dark slate background (#0A0E1A), clean modular containers.'*	**Placement:** Directly below Section 4. **Caption:** *Figure 4.1: Decoupled multi-tier system architecture of AI Teacher platform.*
5. Models Stack **[OPTIONAL]**	**Figure 5.1** **Model Stack Diagram**	**Prompt / Vector Graphic:** *'Clean layered AI tech stack graphic on dark background. Tier 1: Llama-3.3-70B-Versatile (Curriculum & Pedagogy). Tier 2: Llama-3.1-8B-Instant (Fast doubt/quiz eval). Tier 3: Whisper Large v3 (Voice STT). Tier 4: all-MiniLM-L6-v2 (Vector embeddings). Tier 5: Edge-TTS Neural Models. Minimalist tech badges with clean white icons.'*	**Placement:** Under Section 5 table. **Caption:** *Figure 5.1: Layered AI/ML model orchestration and progressive fallback hierarchy.*
6. RAG Pipeline **[REQUIRED]**	**Figure 6.1** **Technical Pipeline Diagram**	**Prompt / Draw.io:** *'Technical data flow diagram showing textbook ingestion. Step 1: Upload (digielec.pdf) ➔ Step 2: Table of Contents Scanner (locates Chapter 2: Binary Codes at page 41) ➔ Step 3: Recursive Character Chunking (1000 char windows, 200 overlap) ➔ Step 4: Sentence Transformers ➔ Step 5: ChromaDB Index ➔ Step 6: Cosine Similarity Top-K Retrieval into System Prompt.'*	**Placement:** Below Section 6. **Caption:** *Figure 6.1: Table of Contents aware vector chunking and strict retrieval grounding pipeline.*

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7. Agent FSM **[REQUIRED]**	**Figure 7.1** **State Machine Diagram**	**Prompt / Mermaid:** *'UML Finite State Machine diagram on dark background. States: INIT ➔ PLANNING ➔ TEACHING_CONCEPT ➔ AWAITING_CHECKPOINT. Conditional paths: IF score >= 0.7 ➔ NEXT_CONCEPT; IF misconception detected ➔ REMEDIATION (Simplified Analogy & Re-eval) ➔ NEXT_CONCEPT. Terminal path: FINAL_ASSESSMENT ➔ COMPLETED. Directional glowing arrows with clear condition text.'*	**Placement:** Below Section 7. **Caption:** *Figure 7.1: Pedagogical Finite State Machine (FSM) governing lesson progression and remediation.*
8. Personas **[REQUIRED]**	**Figure 8.1** **Photorealistic Portraits**	**Prompt 1 (Dr. Sarah):** *'DSLR 85mm f/1.8 photo of friendly 35yo female science educator in modern bright science lab, warm smile, natural skin pores, realistic depth of field, authentic daylight. --no waxy, CGI, plastic, cartoon'* **Prompt 2 (Prof. Aryan):** *'DSLR 85mm f/1.8 photo of distinguished 42yo Indian engineering professor with glasses in university lab, confident analytical expression, natural lighting. --no CGI, glossy'* **Prompt 3 (Coach Maya):** *'DSLR 85mm photo of energetic 28yo female STEM coach in tech workshop, warm smile, authentic expression. --no 3D render'*	**Placement:** Beside Personas subsection. **Caption:** *Figure 8.1: Selectable teacher personalities with custom avatars, vocal timbres, and pedagogical styles.*
9. Checkpoint **[REQUIRED]**	**Figure 9.1** **UI Screenshot**	**Action:** Navigate to **Checkpoints** tab. Capture: 1) The conceptual question card, 2) Student answer input with active microphone recording wave, 3) Evaluation score badge (e.g. 100% or 0%), and 4) Remediation feedback card explaining the simplified mental model.	**Placement:** Under Formative Assessment. **Caption:** *Figure 9.1: Formative micro-checkpoint with speech-to-text response and live rubric remediation.*
9. Diagnostic **[REQUIRED]**	**Figure 9.2** **UI Screenshot**	**Action:** Navigate to **Assessment & Report** tab after completing quiz. Capture: 1) Overall mastery percentage circle (e.g. 85%), 2) Bloom's cognitive taxonomy breakdown bars, 3) Identified strengths & weak areas, and 4) Personalized revision roadmap.	**Placement:** Under Summative Assessment. **Caption:** *Figure 9.2: Comprehensive student diagnostic learning report and revision roadmap.*
10. Multilingual **[REQUIRED]**	**Figure 10.1** **UI Screenshot**	**Action:** During interactive lesson, click the **'Hinglish'** or **'Hindi'** pill on the video player header. Capture the screen showing the active pill, the updated educator script in Hinglish, and Devanagari whiteboard subtitles.	**Placement:** Under Section 10. **Caption:** *Figure 10.1: Dynamic in-lesson multilingual switcher re-narrating the lesson into Hindi/Hinglish.*
11. Voice Pipeline **[OPTIONAL]**	**Figure 11.1** **Audio Architecture Diagram**	**Prompt / Vector Graphic:** *'Bidirectional audio signal processing diagram on dark background. Channel 1 (Input STT): Microphone voice ➔ Web Speech API (Client zero-latency) & Whisper Large v3 (Server multipart). Channel 2 (Output TTS): Educator script text ➔ Edge-TTS Neural Synthesizer (pitch/rate tuned) ➔ WAV audio stream ➔ Video muxing. Clean audio waveform graphics.'*	**Placement:** Under Section 11. **Caption:** *Figure 11.1: Bidirectional voice architecture: Neural Edge-TTS output & Web Speech STT input.*
12. Video Synthesis **[REQUIRED]**	**Figure 12.1** **CV Pipeline Diagram**	**Prompt / Draw.io:** *'Computer vision algorithm pipeline diagram. Step 1: Input Audio (volume envelope & RMS energy extraction over 16 fps slices). Step 2: Haar Cascade Face Landmark Detection (isolating oral cavity & jaw coordinates). Step 3: Affine Warp & Interpolation (cv2.INTER_LINEAR mouth opening & head nod). Step 4: FFmpeg H.264 MP4 muxing (<1.9s latency). Clean engineering schematic.'*	**Placement:** Below Section 12. **Caption:** *Figure 12.1: Sub-2-second audio-driven facial motion & lip-sync synthesis architecture.*
13. API Topology **[OPTIONAL]**	**Figure 13.1** **Cloud Topology Diagram**	**Prompt / Vector Graphic:** *'Microservices cloud integration topology diagram on dark theme. Central FastAPI server node surrounded by secure REST/HTTPS channels to: Groq Cloud API (Llama 3.3 & Whisper), Microsoft Edge-TTS, LangSmith Observability, Pollinations AI, and local ChromaDB/SQLite. Modern AWS/GCP architecture icon style.'*	**Placement:** Under Section 13. **Caption:** *Figure 13.1: External API topology and asynchronous communication channels.*
14. Setup **[OPTIONAL]**	**Figure 14.1** **Terminal Screenshot**	**Action:** Run .\run_fullstack.bat in PowerShell. Capture the console showing: 1) 'Starting AI Teacher FastAPI Backend (Port 8000)', 2) 'Starting Vite React Frontend (Port 5173)', and 3) Clean Uvicorn startup log with WatchFiles reloader active.	**Placement:** Below code block. **Caption:** *Figure 14.1: Automated full-stack execution with separate backend and frontend runners.*

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15. Deployment **[REQUIRED]**	**Figure 15.1 Dashboard Screenshot**	**Action:** Capture the **Render.com** (or Railway) service dashboard showing: 1) Web Service name 'ai-teacher-platform', 2) Green 'Live' status badge, 3) Active environment variables (GROQ_API_KEY, PORT=8000), and 4) The public HTTPS URL link.	**Placement:** Under Section 15. **Caption:** *Figure 15.1: Render.com production deployment dashboard with active HTTPS service.*
16. Trade-Offs **[OPTIONAL]**	**Figure 16.1 Engineering Trade-Off Graph**	**Prompt / Chart:** *'2x2 technical trade-off matrix chart on dark slate canvas. Horizontal axis: Inference Latency (Real-Time <2s vs Batch >30s). Vertical axis: Visual Realism (2D Affine Warp vs 3D NeRF Mesh). Highlighting AI Teacher sweet spot at <2s latency with OpenCV affine lip-sync, solving cloud SadTalker 45s latency. Modern developer documentation graph style.'*	**Placement:** At end of Section 16. **Caption:** *Figure 16.1: Performance trade-off matrix: sub-2-second latency vs heavy 3D mesh rendering.*

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